The Journal of Open Source Software

Athlytics: A Computational Framework for Longitudinal Analysis of Exercise Physiology Metrics from Wearable Sensor Data

Zhiang He[1]

1 Independent researcher

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 15 May 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

The proliferation of wearable sensors and platforms like Strava has generated vast amounts of longitudinal physiological data, offering unprecedented opportunities to investigate dynamic human responses to exercise. However, translating this raw data into scientifically rigorous insights for exercise physiology research faces considerable methodological and computational challenges. `Athlytics` is an R (R Core Team, 2025) package designed as a computational framework to streamline the acquisition, analysis, and visualization of key exercise physiology metrics directly from Strava API data. It enables researchers and practitioners to systematically quantify indicators such as Acute:Chronic Workload Ratio (ACWR) (Gabbett, 2016; Hulin et al., 2016), Efficiency Factor (EF) (Allen et al., 2019), cardiovascular decoupling (Maunder et al., 2021), and track personal bests (PBs). By providing standardized function interfaces adhering to `tidyverse` principles (Wickham et al., 2019), `Athlytics` significantly lowers the technical barrier for conducting complex longitudinal analyses, thereby enhancing the feasibility, efficiency, and reproducibility of quantitative research utilizing widely available physiological sensor data.

Statement of Need

Understanding dynamic physiological responses to exercise is central to exercise science and personalized health (Bourdon et al., 2017). While wearable sensors generate rich data streams (Hicks et al., 2019), established open-source workflows in R for systematically quantifying key physiological indicators (e.g., ACWR, EF, decoupling) from popular platform APIs like Strava are lacking. This often forces researchers to expend considerable effort on custom programming for data retrieval, metric calculation, and longitudinal visualization, limiting analytical scale, efficiency, and reproducibility (Sanders et al., 2017). `Athlytics` addresses this critical bottleneck by providing a dedicated R framework. It seamlessly integrates data acquisition (via the `rStrava` package (Beck et al., 2024)) with the calculation and visualization of these exercise physiology metrics. This empowers broader research applications by enabling researchers to efficiently test hypotheses regarding the dynamic interplay between training stimuli, physiological efficiency, and stress responses using ubiquitous data sources, while carefully considering the necessary approximations for certain composite load metrics.

Key Functionalities

`Athlytics` offers a modular structure, generally featuring paired `calculate_*` and `plot_*` functions for each analytical task, built upon `dplyr` (Wickham et al., 2023), `ggplot2` [Wickham et al. (2025), and other `tidyverse` packages.

- **Data Acquisition and Preprocessing:** Leverages `rStrava` (Beck et al., 2024) for Strava API authentication and data retrieval. Handles date/time operations using `lubridate`

(Grolemund & Wickham, 2011) and rolling calculations with `zoo` (Zeileis & Grothendieck, 2005).

- **ACWR Trend Analysis:**
  - `calculate_acwr`: Computes daily acute (e.g., 7-day) and chronic (e.g., 28-day) rolling loads and their ratio (ACWR) based on user-specified load metrics (duration, distance, or approximated TSS/HRSS).
  - `plot_acwr`: Visualizes the ACWR time series, optionally highlighting risk zones.
- **Load Exposure Analysis:**
  - `calculate_exposure`: Computes daily acute and chronic loads to facilitate a 2D visualization of training state.
  - `plot_exposure`: Generates a scatter plot of acute vs. chronic load, often with ACWR-derived risk zones.
- **Personal Bests (PBs) Tracking:**
  - `calculate_pbs`: Identifies and records best performance times for user-specified distances from Strava activity data (relying on Strava's `best_efforts` field).
  - `plot_pbs`: Visualizes PB progression over time.
- **Efficiency Factor (EF) Analysis:**
  - `calculate_ef`: Computes EF (e.g., average speed/HR or average power/HR) from activity summary data.
  - `plot_ef`: Visualizes EF trends over time, optionally with a smoothing line.
- **Decoupling Analysis:**
  - `calculate_decoupling`: Calculates heart rate/power (or pace/heart rate) decoupling by comparing the efficiency ratio of the first half of an activity to the second half, using detailed activity stream data.
  - `plot_decoupling`: Visualizes decoupling percentages over time.

Comparison with Similar Software

Within the R ecosystem for sports science analytics, `Athlytics` carves out a distinct niche through its integrated analytical workflow specifically designed for Strava API data and a curated set of physiological metrics. While the `rStrava` package (Beck et al., 2024) provides the essential channel for researchers to access Strava data, `Athlytics` builds a critical **analytical layer** upon it, directly transforming raw activity data into physiological insights suitable for longitudinal research. Compared to more broadly functional R packages like `trackeR` (Frick & Kosmidis, 2017)—which excels in handling diverse local sports tracking files (e.g., .tcx, .gpx) and offering general-purpose sports science analytics—`Athlytics` focuses on the direct utilization of Strava API data and provides **ready-to-use modules** for core physiological indicators of interest to researchers (such as ACWR, EF, and decoupling). This design obviates the need for researchers working with Strava data to undertake cumbersome data integration and workflow concatenation, as might be required when using multiple single-function R packages (e.g., a dedicated package for ACWR calculation like `ACWR` (Rogers, 2022), or custom scripts). `Athlytics` offers a **coherent framework from data acquisition to the calculation and visualization of multi-dimensional physiological metrics**. Adhering to `tidyverse` design principles, it not only lowers the technical barrier for researchers conducting such specific analyses but, more importantly, significantly enhances the **reproducibility and methodological transparency** of exercise physiology research based on wearable device data by providing an open-source, standardized computational environment, directly addressing the core values of scientific software.

Illustrative Example

The following example demonstrates how to generate an ACWR trend plot for running activities based on duration.

```
library(Athlytics)
```

```r
# Assume 'stoken' is a valid Strava token object.
# Example: stoken <- rStrava::strava_oauth(YOUR_APP_NAME, YOUR_CLIENT_ID, YOUR_CLIENT_SE

# 1. Plot ACWR Trend
acwr_plot <- plot_acwr(
  stoken = stoken,
  activity_type = "Run",
  load_metric = "duration_mins",
  acute_period = 7,
  chronic_period = 28
)
# print(acwr_plot)

# 2. Plot Load Exposure
exposure_plot <- plot_exposure(
  stoken = stoken,
  activity_type = "Ride",
  load_metric = "tss", # Requires user_ftp if using "tss"
  user_ftp = 280,      # Example FTP value
  acute_period = 7,
  chronic_period = 28
)
# print(exposure_plot)

# 3. Plot Efficiency Factor (EF) Trend
ef_plot <- plot_ef(
  stoken = stoken,
  activity_type = c("Run", "Ride"),
  ef_metric = "Pace_HR", # Or "Power_HR"
  min_duration_mins = 20
)
# print(ef_plot)

# 4. Plot Personal Bests (PBs) Progression
pbs_plot <- plot_pbs(
  stoken = stoken,
  distance_meters = c(1000, 5000, 10000), # Example distances
  activity_type = "Run"
)
# print(pbs_plot)

# 5. Plot Decoupling Trend
# Note: This function can be slow as it fetches detailed activity streams.
decoupling_plot <- plot_decoupling(
  stoken = stoken,
  activity_type = "Run",
  decouple_metric = "Pace_HR", # Or "Power_HR"
  max_activities = 20 # Limit activities for example speed
)
# print(decoupling_plot)
```


Acknowledgments

The development of Athlytics (He, 2025b), which is now available from the Comprehensive R Archive Network (CRAN) (https://CRAN.R-project.org/package=Athlytics), relied upon

the R programming language (R Core Team, 2025) and benefited from numerous open-source R packages, including rStrava (Beck et al., 2024), ggplot2 [Wickham et al. (2025), dplyr (Wickham et al., 2023), tidyr (Wickham et al., 2024), lubridate (Grolemund & Wickham, 2011), zoo (Zeileis & Grothendieck, 2005), purrr (Wickham & Henry, 2025), and rlang (Henry & Wickham, 2025). Access to data was made possible by the Strava API. We also acknowledge the preprint of this work on bioRxiv (He, 2025a).

Allen, H., Coggan, A. R., & McGregor, S. (2019). *Training and Racing with a Power Meter* (3rd ed.). VeloPress. ISBN: 9781937715939

Beck, M. W., Villarroel, P., Padfield, D., Gaborini, L., & Maltzahn, N. von. (2024). *rStrava: Access the 'Strava' API*. https://doi.org/10.32614/CRAN.package.rStrava

Bourdon, P. C., Cardinale, M., Murray, A., Gastin, P., Kellmann, M., Varley, M. C., Gabbett, T. J., Coutts, A. J., Burgess, D. J., Gregson, W., & Cable, N. T. (2017). Monitoring Athlete Training Loads: Consensus Statement. *International Journal of Sports Physiology and Performance*, *12*(Suppl 2), S2-161-S2-170. https://doi.org/10.1123/ijspp.2017-0208

Frick, H., & Kosmidis, I. (2017). trackeR: Infrastructure for Running and Cycling Data from GPS-Enabled Tracking Devices in R. *Journal of Statistical Software*, *82*(7), 1–29. https://doi.org/10.18637/jss.v082.i07

Gabbett, T. J. (2016). The training—injury prevention paradox: should athletes be training smarter and harder? *British Journal of Sports Medicine*, *50*(5), 273–280. https://doi.org/10.1136/bjsports-2015-095788

Grolemund, G., & Wickham, H. (2011). Dates and Times Made Easy with lubridate. *Journal of Statistical Software*, *40*(3), 1–25. https://doi.org/10.18637/jss.v040.i03

He, Z. (2025a). Athlytics: A Computational Framework for Longitudinal Analysis of Exercise Physiology Metrics from Wearable Sensor Data. *bioRxiv*. https://doi.org/10.1101/2025.05.01.651597

He, Z. (2025b). *Athlytics: Advanced Sports Performance Analysis for 'Strava' Data*. https://doi.org/10.32614/CRAN.package.Athlytics

Henry, L., & Wickham, H. (2025). *rlang: Functions for Base Types and Core R and 'Tidyverse' Features*. https://doi.org/10.32614/CRAN.package.rlang

Hicks, J. L., Althoff, T., Sosic, R., Kuhar, P., Bostjancic, B., King, A. C., Leskovec, J., & Delp, S. L. (2019). Best practices for analyzing large-scale health data from wearables and smartphone apps. *Npj Digital Medicine*, *2*(1), 45. https://doi.org/10.1038/s41746-019-0121-1

Hulin, B. T., Gabbett, T. J., Lawson, D. W., Caputi, P., & Sampson, J. A. (2016). The acute: chronic workload ratio predicts injury: high chronic workload may decrease injury risk in elite rugby league players. *British Journal of Sports Medicine*, *50*(4), 231–236. https://doi.org/10.1136/bjsports-2015-094817

Maunder, E., Seiler, S., Mildenhall, M. J., Kilding, A. E., & Plews, D. J. (2021). The Importance of 'Durability' in the Physiological Profiling of Endurance Athletes. *Sports Medicine*, *51*(8), 1619–1628. https://doi.org/10.1007/s40279-021-01459-0

R Core Team. (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Rogers, N. (2022). *ACWR: Functions to Calculate the Acute:Chronic Workload Ratio (ACWR)*. https://doi.org/10.32614/CRAN.package.ACWR

Sanders, D., Abt, G., Hesselink, M. K. C., Myers, T., & Akubat, I. (2017). Methods of Monitoring Training Load and Their Relationships to Changes in Fitness and Performance in Competitive Road Cyclists. *International Journal of Sports Physiology and Performance*, *12*(5), 668–675. https://doi.org/10.1123/ijspp.2016-0454

DRAFT

Wickham, H., Averick, M., Bryan, J., Chang, W., D'Agostino McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Wickham, H., Chang, W., Henry, L., Pedersen, T. L., Takahashi, K., Wilke, C., Woo, K., Yutani, H., & Dunnington, D. (2025). *ggplot2: Create elegant data visualisations using the grammar of graphics*. https://doi.org/10.32614/CRAN.package.ggplot2

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *dplyr: A Grammar of Data Manipulation*. https://doi.org/10.32614/CRAN.package.dplyr

Wickham, H., & Henry, L. (2025). *purrr: Functional Programming Tools*. https://doi.org/10.32614/CRAN.package.purrr

Wickham, H., Vaughan, D., & Girlich, M. (2024). *tidyr: Tidy Messy Data*. https://doi.org/10.32614/CRAN.package.tidyr

Zeileis, A., & Grothendieck, G. (2005). zoo: S3 Infrastructure for Regular and Irregular Time Series. *Journal of Statistical Software*, *14*(6), 1–27. https://doi.org/10.18637/jss.v014.i06

DRAFT